valuations are represented by the ten-year average of real earnings divided by the real stock market index value at the beginning of the retirement year (E10/P). This is just the inverse of Shiller's PE calculation. When this smoothed earnings yield is low, the stock market tends to be overvalued as a result of a recent run-up in stock prices. Stock price appreciation helps the worker reach her wealth accumulation goal with a lower savings rate. At the same time, a highly valued stock market at the retirement date—as represented by the low smoothed earnings yield—also correlates with a lower subsequent MWR for the new retiree. This point is covered more thoroughly in the previous section.

Now we are ready to integrate the working and retirement phases to determine the savings rate needed to finance the planned retirement expenditures for rolling sixty-year periods from the data. The baseline individual wishes to withdraw an inflation-adjusted 50 percent of her final salary from her investment portfolio at the beginning of each year for thirty years. Prior to retiring, she earns a constant real salary over thirty years, and her objective is to determine the minimum necessary savings rate to be able to finance her desired retirement expenditures. Her asset allocation during the entire sixty-year period is 50/50 for stocks and short-term fixed-income assets.

Exhibit 5.11

Minimum Necessary Savings under Isolated and Integrated Approaches

For 50/50 Asset Allocation, 30-Year Work Period, 30-Year Retirement Period, 50% Replacement Rate